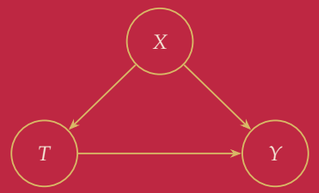
Causes and effects are discoverable,
not by reason but by experience.

David Hume

Every causal question is a question about something we never see: the outcome that would have occurred under a different treatment. Double machine learning answers it with orthogonal scores, sample splitting and asymptotic confidence intervals. This book answers it with probability.

Bayesian Causality builds causal inference from the ground up as a problem of models and accounting. Treatment effects become posterior quantities, confounding becomes a statement about priors and likelihoods, and uncertainty is carried through every step rather than patched on at the end. Along the way the authors decompose the error of double machine learning, show where its guarantees come from and where they break, and develop Bayesian alternatives that remain coherent in finite samples, with applications from policy evaluation to the effects of mergers.

Written for statisticians, economists and data scientists, and for anyone who must decide what caused what.



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Cover: Vincent van Gogh, The Starry Night, *1889.*
The forces are invisible; we see only the swirl they leave.

Bayesian Causality

Polson • Sokolov

Bayesian Causality

Models, Accounting and the Bayesian Alternative to Double Machine Learning



Nick Polson and Vadim Sokolov